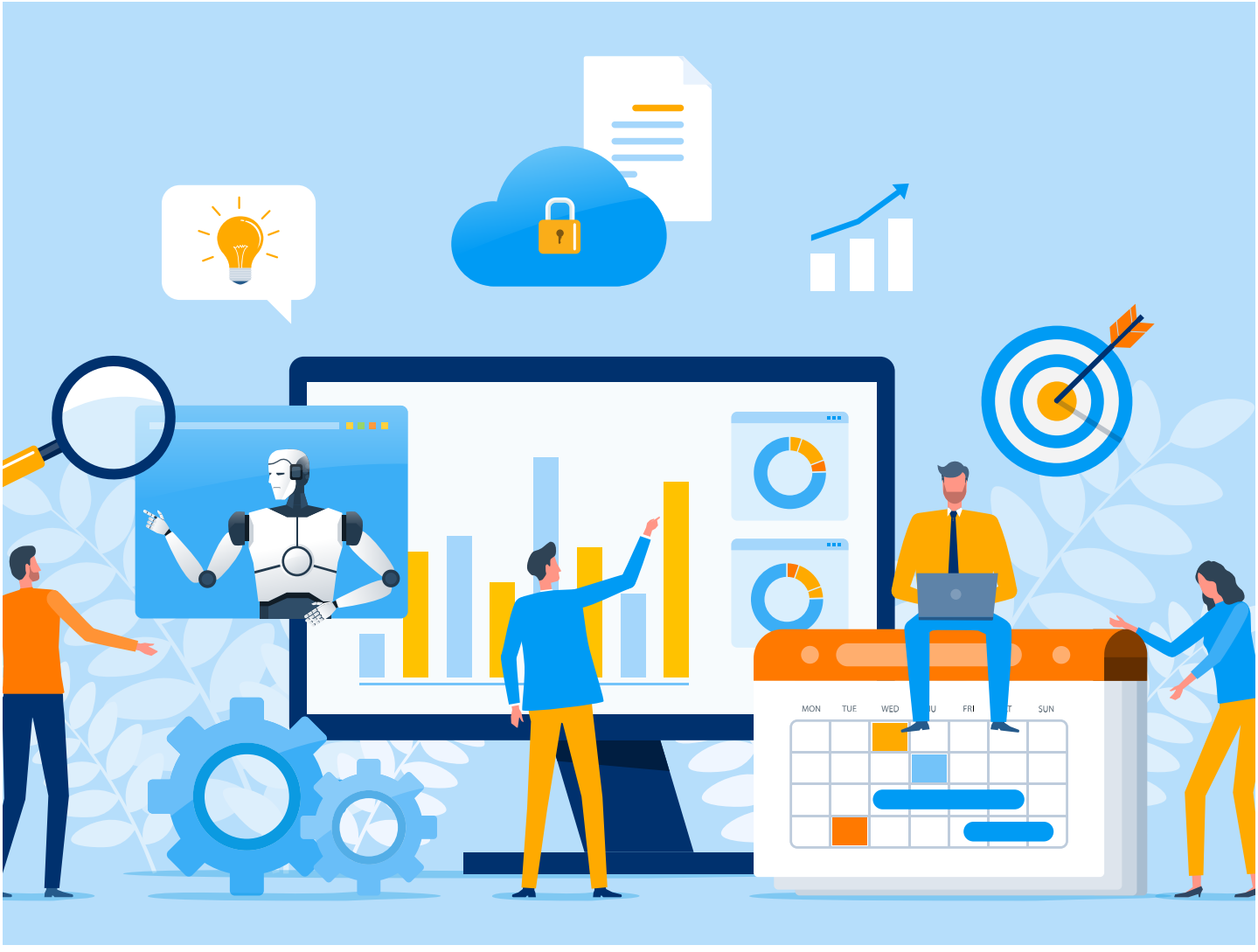
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Insights

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Predictive modeling with AI can revolutionize how organizations use everyday business data.

Bringing predictive analytics to the agentic AI era



Foreword

For most of the past decade, the competitive question in analytics was who could predict better. That question is largely settled. The harder divide now separating leaders from everyone else is who can act on a prediction autonomously, at scale, in real time, and without drifting from business intent.

This is where most organizations quietly stall, and the reason is rarely the model. A system can correctly predict that a customer is likely to churn or that a claim is worth a certain amount, and still produce the wrong decision because it does not understand the process it is supposed to operate within. That knowledge does not live in the data. It lives in the operational depth of people who have run the process thousands of times.

Take collections. A model can rank customers by likelihood of payment. But the decision that actually matters is more delicate. For example, one customer just had a payment fail for the third time, they are in a jurisdiction with specific hardship protections, the last agent who called them noted a recent bereavement, and the right move is not another payment reminder but a pause and a different tone entirely. A frontline specialist who has handled ten thousand of these conversations knows that instinctively. They know the ordinary path, and, more importantly, the hundred corner cases where that path is exactly wrong. The same logic applies in insurance claims assessment, fraud detection, and healthcare demand forecasting. The domain changes, but the problem does not.

That accumulated judgment is the real asset. The routine cases, the difficult ones, and the rare exceptions are precisely the material we use to train AI agents — so the system learns not just the typical decision but the full distribution of situations a human expert has learned to navigate. An agent trained only on the common path fails quietly when it meets the edge case, and at scale, that failure compounds.

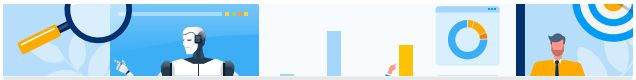
This report moves past the comfortable consensus that predictive AI is powerful, and onto the harder question: what does it take to let these systems act? The answer is the depth of process knowledge, corner cases, and all that gets encoded into the agents themselves. And the gap between organizations that have that depth and those that don't is widening every quarter.

Sebastian Cubela
Global Head of Data and AI Partnerships, TP

In 2026, the question for enterprise AI is no longer whether predictive models can outperform statistical forecasts – that argument is settled. The big question now is how to enable predictive systems to act on their own conclusions without drifting from business intent. The frontier has moved from prediction to autonomous decision making, and the gap between leaders and laggards is widening accordingly.

"Enterprises are done with a backward-looking point of view; they want to be more forward-thinking," says Vishal Gupta, partner at research firm Everest Group.

Intelligent analytics, powered by technologies like deep learning and generative AI, are making this possible. Real-time training allows AI to evolve continuously instead of waiting for quarterly refreshes. In addition, the data that newer predictive engines rely upon has expanded to encompass not just neat, numerical records but also messy, unstructured sources of insight-rich interactions. As a result, AI-powered analytics are moving enterprises from passive hindsight to pragmatic foresight.



The value of the global predictive analytics market

2025 $22.22 billion

2026 $27.56 billion

2034 $116.65 billion

Source: Compiled by MIT Technology Review Insights based on data from Fortune Business Insights, 2026

Key takeaways



1. Predictive analytics has shifted from static forecasts to living systems – ingesting unstructured data, updating in real time, and surfacing insight through natural-language interfaces.
2. Intelligent analytics have wide-ranging applications across industries including healthcare, financial services, and manufacturing. But challenges around data readiness, strategic alignment, talent, and maintaining predictive models through continuous improvement persist.
3. Few enterprises have the in-house breadth to orchestrate the full stack – technology, talent, governance, and industry-specific use cases. Partnering is becoming the default path to scaled AI, not a stopgap.

How AI is reshaping predictive analytics

AI takes predictive analytics – a broad discipline that includes predictive modeling, data prep, analysis workflows, interpretation of results, and decision-making applications – to new heights. "In many ways I think the word 'analytics' is giving way to AI," says Gupta. "Everything is becoming AI."

Several technological breakthroughs are to thank. First is the ability of generative AI systems to ingest unstructured data at scale. "[AI systems] can now process text, voice, image, and streaming data, which means real-time integration of inputs like sensors, customer interaction data, and social media interactions," explains Gupta, noting that these previously untapped goldmines often provide far richer and "situationally aware" intel than a standard customer satisfaction score (CSAT) or net promoter score (NPS).

The second breakthrough is the move from static, infrequently-updated models to real-time intelligence. Instead of waiting weeks for an analyst to refresh a dashboard, AI models now learn continuously from live data streams.

A third development pertains to user interfaces. Past iterations of predictive technologies were often clunky for the average frontline employee with minimal data analysis expertise to navigate. “The interface was just not [practical],” says Pedro Amorim, co-founder of consulting firm LTP Labs and a professor teaching data-driven decision making at the University of Porto in Portugal. Users can engage with today’s generative AI systems via intuitive natural-language commands.

Classification of data, too, has become more streamlined. Danny Kuivenhoven, chief technology officer for EMEA and APAC at TP, gives an example in the context of customer interactions. “When classification becomes consistent at scale, something bigger happens – you stop looking at customer interactions as conversations and start seeing them as a structured signal you can actually act on. That’s the moment predictive analytics changes from a reporting tool into an operating system for the business,” says Kuivenhoven.

At scale, this capability also becomes a powerful lens for operational insight. We can get almost a FinOps-level operation showing a breakdown of volumes and associated costs. For instance, ‘This month you spent $250,000 on returns, and this month we spent $100,000 on payment questions,’” Kuivenhoven continues. Such findings can more accurately feed predictive models and help organizations identify emerging cost drivers or proactively adjust staffing and workflows.

The final piece of the puzzle, notes Gupta, is integration. More and more organizations are now embedding predictive capabilities directly into automated workflows – and the rise of AI agents will almost certainly accelerate this trend.

“Enterprises are no longer treating predictive analytics and automation as separate initiatives. Instead, they’re creating closed-loop decision systems. Predictive analytics uses the data, be it real-time, structured, or unstructured, to create a set of insights based on which agentic AI can make certain autonomous decisions.”

Vishal Gupta, Partner, Everest Group

How multimodal data changes predictive modeling



Danny Kuivenhoven, chief technology officer for EMEA and APAC at TP, provides an example of how multimodal data can add new dimensions to predictive insight: Smart video systems that blend predictive analytics with computer vision and generative AI. “We can see a problem through users’ mobile phone cameras, and now we can actually do something directly with the images,” he says.

In car insurance claims, for example, this capability allows organizations to move beyond historical averages. Instead, AI systems can analyze visual inputs alongside structured information to assess damage in real time. “We can create a predictive model based on the type of car, the age of the vehicle, and the severity of the damage, and we can estimate the potential value of that claim,” explains Kuivenhoven.

Despite these advances, human oversight remains essential, particularly when models are making high-stakes financial calls. Visual data can be ambiguous, and edge cases are inevitable. “That’s why it’s important to have human annotators review and correct the algorithm as needed,” he says.



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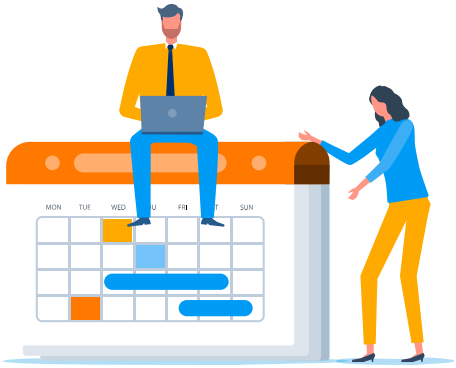
Wide-ranging industry applications

Kuivenhoven sees this shift as a fundamental expansion of predictive analytics' role. "Previously, predictive capabilities focused primarily on what was happening next, but now intelligent prediction is meeting action." He points to several recent client engagements:

- In B2B sales, a multinational food and beverage services provider hoped to use AI to read buying patterns more accurately and better anticipate churn. To address this need, TP built out propensity and segmentation models based on three years of interaction data, grouping accounts by order size and frequency. Internal data from the company shows results of the initiative: 11% more sales tickets, a 27% rise in average revenue per ticket, 70% more accurate churn forecasts, and a 41% gain in NPS. At a US technology company, the same interaction-analytics approach automated 100% of quality audits and cut repeat calls by roughly a quarter, which from internal TP data shows a 38% lift in sales conversion.

- Collections, in which AI models decide not only who to contact, but when and how often based on their prior behavior, is another promising area of application. To put this into practice for a US-based retailer trying to outperform a rival vendor on past-due accounts before write-off, TP scored each account for likelihood of payment. Internal TP data shows an 11.2% higher liquidation rate than the competing vendor, plus $1.8 million in additional recoveries over nine months.

- Hiring is a newer use case for predictive analytics. To assist a hospitality company in the UK struggling with sales targets and early turnover, TP trained machine learning models on psychometric results and the profiles of top-performing agents to identify the factors predicting success, then used the models to inform a new hiring campaign. Kuivenhoven says agents selected this way raised average sales per ticket by 55% and were 40% less likely to leave the organization.



"Whereas in the past, predictive capabilities focused primarily on the next best offer, next best action, propensity of payment, best time to reach customers, etc., we're now extending that into a complete value chain."

Danny Kuivenhoven, Chief Technology Officer, EMEA and APAC, TP

Building the foundation for AI orchestration at scale

Despite the hype around AI adoption, hurdles remain. And many of the real challenges are organizational rather than purely technical. Data readiness, for example, is an ongoing process that many companies misjudge as a one-time effort. "Your data is never going to be in an 'optimal' state. Knowing that, what you need to ask is, 'Which data do I really need to get ready?'" says Amorim. "For that, you need to answer another question first: 'What are the business cases that I really want to solve?'"

Equally important is ensuring strategic alignment before tacking on technological bells and whistles. Gupta emphasizes the importance of outlining clear key performance indicators tied to business outcomes ahead of any predictive AI implementations. Without such alignment, analytics projects risk becoming proof-of-concept exercises that never translate into return on investment.

A commitment to quality assurance is another core tenet. Kuivenhoven notes that a focus on continuous improvement is an often-overlooked factor in maintaining predictive models' potency. A commitment to quality assurance will be even more critical as AI starts to take actions on its own. "The biggest flaw I see today in autonomous systems is that there's no form of continuous improvement," Kuivenhoven says. "Many vendors simply conduct periodical reviews on quality and intervene [only when it has obviously degraded]." This can translate to growing erosion of user trust or decisions that quietly drift away from business objectives.

The next challenge is talent. The expanding AI ecosystem covers everything from generative and descriptive AI to prescriptive and simulation-based approaches, and organizations need people who can navigate those overlapping ideas. "You really need to know at least a little bit about all these different AI technologies, because otherwise you might be using a hammer to nail in a screw," says Amorim.

Finally, sustainable impact requires cross-functional collaboration. Gupta cautions against siloed IT-led projects, stressing that the talent/team structure for deploying AI needs to be a holistic process. "It cannot just be the tech teams; it cannot just be the domain teams," he emphasizes.

For many organizations, the breadth of expertise required to get the most out of AI's predictive capabilities makes a strong case for partnering with a service provider that has already navigated these complexities. Such partners can offer what might be called "AI orchestration." a multifaceted approach that combines the right technology, people, and proven, industry-specific use cases to optimize results.

Responsible implementation and governance

Disciplined data governance is another cornerstone of effective predictive AI adoption, especially for industries dealing with sensitive data. "In the market today, there's a big maturity gap with AI. Companies that are low on that scale and experimenting with entry-level AI are often relying heavily on the guardrailing that the major large language model providers include," says Kuivenhoven. As organizations move toward more specialized use cases – and begin training or fine-tuning models using proprietary data – that level of built-in protection is no longer sufficient.

Synthetic data as a catalyst for predictive modeling innovation

Clean and compliant data is perhaps the most important component of effective predictive modeling, but this precious resource can be both costly to obtain and fraught with privacy concerns. In regulated industries in particular, customer interaction records or health data may be too sensitive to use freely. In other cases like simulating an anomalous event or disruption, the right data simply might not exist in sufficient volume.

This is where the value of synthetic data—artificially generated datasets that mirror the statistical properties of real-world information—is starting to become clear. Such datasets help address the twofold challenge of data scarcity and data sensitivity.

Synthetic data can be created and refined in a few different ways:

- **Algorithms generate new records based on patterns learned from actual datasets**
- **Simulation engines replicate real-world processes, such as traffic flows or patient care pathways, to create plausible test scenarios**
- **Data augmentation techniques add variation to existing records (e.g., altering geolocations or word choices in customer transcripts) to broaden model training sets**

At TP, synthetic data is standard practice for development and experimentation. "Anything new that we build, we start with synthetic data, test that it works, and then slowly move into real live data to see if we're getting the same results," says Kuivenhoven. The approach also supports guardrailing and compliance testing. Synthetic datasets make it easier to verify that monitoring systems trigger the right alerts before deploying in production environments.

Still, generating high-quality synthetic data requires careful methodology to avoid introducing new biases or simply replicating the blind spots of existing models. For organizations without deep data science expertise, partnering with a provider experienced in these techniques can accelerate experimentation while minimizing risk.

Effective governance begins with robust redaction and anonymization practices to ensure customer information is protected throughout the analytics lifecycle. Truly responsible governance also requires thinking beyond the back-office; it necessitates direct client involvement to establish what data can be collected, how it will be used, and under what safeguards. In regulated industries such as healthcare and finance, these guardrails often define whether an AI initiative can scale beyond pilot programs.

The tumultuous regulatory landscape adds urgency to governance efforts. Gupta believes pressure will only mount as more laws like the EU Artificial Intelligence Act take effect. "In the coming months and years, the EU's AI regulations will come into full force. Enterprises are increasingly considering this in their technology systems, but also in their governance processes. Who can access what data? Where is this data being used, and in what kind of models?" he says.

In terms of early best practices, many enterprises are deploying "walled gardens" for AI, where proprietary or sensitive data is isolated and processed within secure, controlled environments. Documentation and testing protocols are becoming standard, ensuring models can be explained and audited.

Moving forward, the emphasis will likely be on making governance frameworks more proactive and scalable. Synthetic data and advanced cleaning techniques will help reduce bias and limit reliance on raw personal data while preserving analytical accuracy. Meanwhile, human-in-the-loop approaches will continue to provide oversight at critical junctures.

Looking forward

AI is, of course, still not an oracle; it cannot guarantee business outcomes. But with every leap forward in model design and computational power, its ability to deliver timely, actionable insights grows more impressive. In the context of predictive analytics, this means the days of linear projections are coming to an end. In its place, an era of reliable, living systems is taking shape.

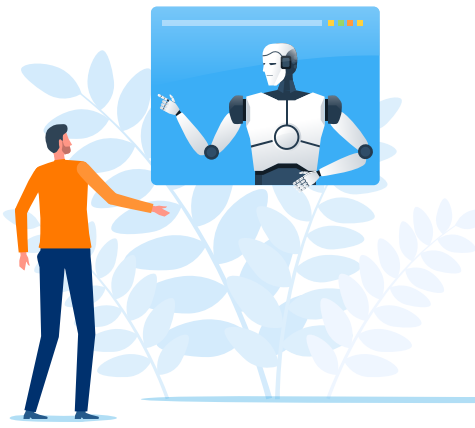
Success with this new paradigm will depend on the balance organizations strike between machine intelligence and human judgment. Gupta observes that we are on the cusp of a major cultural shift, one in which both AI systems and human domain experts will guide how effective new predictive models can be.

"It's a one-way street now with AI, but one thing is very clear: The future is human-AI collaboration," he says. "There will be tasks that will be completely taken over by AI and tasks that will be completely done by humans, but the biggest chunk of all activities will involve humans-plus-AI coming together in varying degrees. That's the future we are headed for, and the sooner enterprises and organizations realize it, the faster they can gain competitive advantage."

That collaborative ethos must also shape how organizations think about outside expertise. As the ecosystem continues to evolve, few organizations will have the internal resources to keep pace with all the moving parts of AI orchestration on their own. Partnering with providers experienced in these complexities will be a defining factor for realizing the technology's true potential.

"Your data is never going to be in an 'optimal' state. Knowing that, what you need to ask is, 'Which data do I really need to get ready? For that, you need to answer another question first: 'What are the business cases that I really want to solve?'"

Pedro Amorim, Co-Founder, LTP Labs, and Professor, University of Porto



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From the sponsor

TP brings nearly five decades of experience running critical customer and business operations. We help the world's leading brands move beyond isolated AI pilots and scale AI-orchestrated operations by combining technology, hybrid workforces, process rigor, and deep industry expertise to deliver measurable outcomes through:

- A comprehensive AI-powered service portfolio covering the full customer lifecycle: front-office customer service, RaaS (revenue-as-a-service), debt collection, trust and safety, and back-office operations, turning interaction data into structured signals that drive better outcomes across every function.

- End-to-end AI data services spanning data engineering, multimodal annotation, AI safety and governance, and advanced analytics, giving organizations the foundation to build, deploy, and continuously improve predictive models at scale, including synthetic data capabilities to accelerate experimentation.

- Industry-specific services for regulated sectors, including real-time fraud detection and AI-assisted claims assessment in financial services and insurance, patient demand forecasting in healthcare, and the data governance frameworks that regulated AI deployments require.

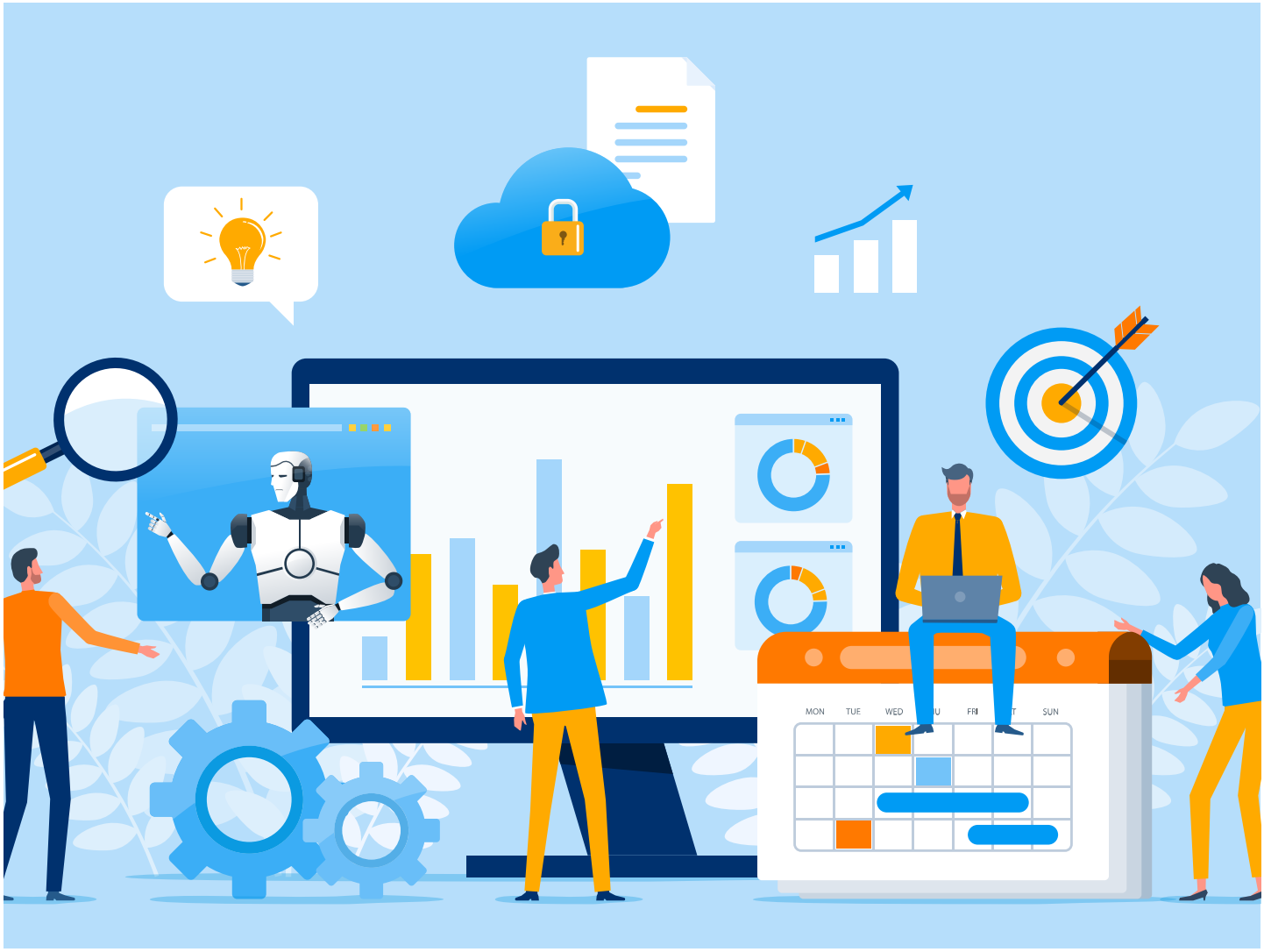
- A suite of plug-and-play solutions that combine advanced AI capabilities with process excellence, integrating within a company's existing technology ecosystem and enabling organizations to deploy targeted predictive capabilities without rebuilding existing infrastructure.



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